

The support of the available bit rate bearer capability using virtual source/destination concepts

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The available bitrate (ABR) bearer capability is described and contrasted with other bearer capabilities of the ATM layer. The significant benefit of ABR is seen as its ability to multiplex very high bit rate sources, where just a few active sources are sufficient to achieve high link utilisation. This paper describes a proposal for adding ABR controls to existing ATM switches so that different bandwidth allocation policies can be tried out. A detailed description is given of an ABR private virtual path (VP) and it is shown how this can be used as a basis for supporting a LAN emulation service, incorporating switched virtual circuits (SVCs). The so-called virtual source/destination concept is used as the basis of the proposal, with some novel additions described to cater for the complexities of fair bandwidth allocation. Conclusions are drawn which emphasise the rapid development of new ideas and the need for flexibility when introducing ABR.

1. Introduction

At the ATM layer, a number of bearer capabilities have been identified [1]:

- the deterministic bit rate (DBR) bearer capability, which can be used for the support of circuit emulation and for constant bit rate connections, provides quality of service commitments for cell loss, cell transfer delay and end-to-end cell delay variation (CDV),
- the statistical bit rate (SBR) bearer capability, which could be used to support variable bit rate connections that have specific requirements both in terms of a peak cell rate (PCR) and a sustained cell rate (SCR), has a quality of service (QOS) commitment in terms of cell loss ratio, but, typically, the permissible cell loss ratio is much larger than for DBR — it is not clear that this capability could provide any commitment in terms of end-to-end CDV,
- the available bit rate (ABR) bearer capability, used to support non-real time variable bit rate connections requiring very low cell losses, has a QOS commitment only in terms of cell loss ratio where the permissible level is similar to that for DBR — there are no commitments on CDV or cell transfer delay,
- the ATM block transfer (ABT) capability is a resource allocation in the network which is block oriented (a block being defined as a number of user cells between two successive block-indicating resource management cells) — blocks are discarded or transferred with very low cell losses and with low CDV.

There is clearly some overlap in the SBR, ABR, and ABT capabilities and it may be that not all of them will be realised in a given ATM network. One possible way of looking at these different capabilities would be to list traffic types and performance objectives where each would operate well (however, this does not imply that they could not be used in other scenarios):

- SBR applies very well to non-real time variable bit rate applications where the sending rate is much smaller than physical link capacities and, where sustained, high-link utilisation is not the aim,
- ABR applies well to non-real time variable bit rate applications where the sending rate of just a few end systems would be enough to occupy all of the capacity of a physical link, and where there is an aim to keep utilisation high (e.g. on an undersea cable),
- ABT applies well to applications which cannot accurately determine their SCR and which may want some QOS commitments in terms of end-to-end CDV and where a sustained high-link utilisation is not the aim.

Keeping link utilisation high implies a control regime where the sending rates of end systems can be altered dynamically. This allows a few connections to use all of the available bandwidth at one moment while allowing re-arrangement of sending rates when further connections are established.

The control of sending rate could be through network commands or simply through reactions to data loss or

excessive delays by the end-system protocols (e.g. TCP). The advantage of controlling the sending rate through network commands is that it allows different bandwidth allocation policies to be implemented. This is much harder to achieve through end-system protocols alone. Furthermore, if it works well, the network-oriented controls can prevent end-systems from continually dropping back to low sending rates because of congestion. This may be achieved by minimising the frequency of changes, and the cell loss ratio, while still keeping sending rates as high as possible.

This paper will look in more detail at ABR and how it may be introduced as a new capability on an existing ATM network platform.

2. Outline of the source and destination end system behaviour for ABR

2.1 Basic concepts

The following description is based upon the current specification that is being developed within the ATM Forum and ITU-T SG 13. This specification has undergone many revisions in the last year and the final form of the standard behaviour of the source end system (SES) and destination end system (DES) may still differ from the details described here. Nevertheless it is included because it provides an introduction to the basic principles of rate-based control.

Referring to Fig 1, an SES is shown connected to a DES via several switches. For simplicity, only a unidirectional flow of data is assumed (from the SES to the DES).

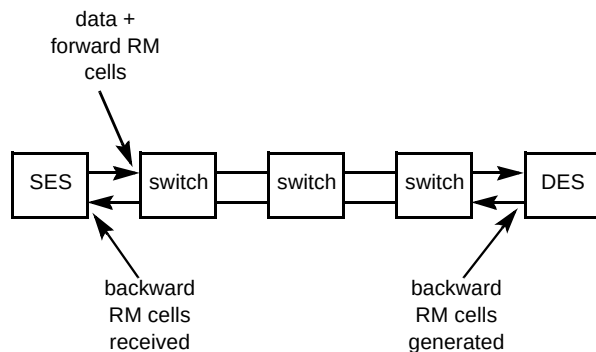


Fig 1 The ABR control loop for a unidirectional data flow.

The fundamental concept is that the SES rate of sending is modified through feedback received on the control loop that extends from the SES, through all the switches to the DES, and back. This control loop allows any of the switches, or the DES, to indicate to the SES that congestion conditions exist (typically this means that a buffer used by this data flow is more than half filled). It further allows the switches and DES to resolve what is the current maximum acceptable sending rate if they have the capability of setting an explicit rate.

All control indications are transported in resource management (RM) cells. An RM cell is generated regularly by the SES and traverses the same route as the data cells. Any switch along the route may overwrite certain fields of the RM cell to initiate a change of sending rate. In the return direction, a backward RM cell is generated by the DES, taking account of indications received on the forward path. The DES may also require a change in the sending rate and resolves any conflict between its own requirements and the indications received in the forward direction. Finally, there is a further opportunity for the switches to amend the contents of the RM cells as they traverse the backward path.

An oversimplified view of this process is that the SES is generating a regular set of requests or tests of the network looking for more bandwidth. The tests are the RM cells, and if their contents remain unmodified, the test is deemed to be successful. In this case the sending rate will be increased by an amount which is part of the standardised procedure — otherwise it is decreased, again in a standardised way.

According to this simple view of the process, an SES will continually increase its sending rate using a series of step changes in rate; this process continues until the switches or the DES are unable to accept a further increase, fulfilling one of the central objectives of driving up the utilisation of otherwise spare network resources.

A desirable feature is that the length of time it takes the SES to achieve its highest allowable rate is small compared to the typical length of time the SES will be actively sending data. However, a limiting factor is the round-trip delay of the control loop. For a wide area network (WAN), this may be tens of milliseconds. Another limiting factor is that, during the first round-trip delay period, it may be necessary to limit the sending rate to a quite low value so as to be cautious, given that no opportunity for feedback has yet been possible. All of this suggests that the parameters of ABR need to be tuned to the network, particularly whether it is wide area or local area, so as to get optimum performance.

2.2 Formal description of the SES, DES and switch behaviour for ABR

The SES behaviour is partly described formally by the following statements¹ (for brevity, some of the description is omitted where it is not further discussed here, but the full description can be found in Sathaye [2]).

¹ The description includes some fixed parameters, namely:
 Nrm — controlling the frequency of sending an RM cell;
 RDF — a fixed rate decrease factor dividing the allowed current rate when congestion is indicated;
 AIR — a factor of the fixed additive increase in rate when no congestion is indicated;
 Xrm — a fixed cell credit value used with XDF;
 XDF — a multiplicative decrease factor operated when no feedback has been received following the expiry of the credit Xrm.

- The value of the allowed cell rate (ACR) shall never exceed the peak cell rate (PCR), nor shall it ever be less than the minimum cell rate (MCR). The SES may always send in-band (i.e. CLP = 0) cells at a rate less than or equal to ACR.
- Before an SES sends its first cell it shall set ACR to, at most, the initial cell rate (ICR). The first in-band cell sent shall be a forward RM cell.
- After the first in-band forward RM cell, in-band cells shall be sent in the following order:
 - the next in-band cell shall be a forward RM cell if, and only if, the last in-band forward RM cell was sent, either at least two in-band cells have been sent and at least T_{rm} seconds have elapsed, or $N_{rm}-1$ in-band cells have been sent,
 - the next in-band cell shall be a backward RM cell if the above condition is not met, if a backward RM cell is waiting for transmission, and if either no backward in-band RM cell has been sent since the last in-band forward RM cell, or no data cell is waiting for transmission,
 - the next in-band cell shall be a data cell if neither of the above two conditions is met, and if a data cell is waiting for transmission.
- Before sending an in-band forward RM cell, if at least X_{rm} in-band forward RM cells have been sent since the last backward RM cell was received, then ACR shall be reduced by at least $ACR \cdot X_{DF}$ unless that reduction would result in a rate below MCR, in which case ACR shall be set to MCR.
- When a backward RM cell is received with congestion indication (CI) = 1, then ACR shall be reduced by at least $ACR \cdot N_{rm}/R_{DF}$, unless that reduction would result in a rate below MCR, in which case ACR shall be set to MCR. If the backward RM cell has both CI = 0 and NI = 0 then the ACR may be increased by no more than $AIR \cdot N_{rm}$ to a rate no greater than PCR.
- When any backward RM cell is received, and after ACR is adjusted according to the previous statement, if ACR is greater than the explicit rate (ER) from the RM cell, then ACR shall be reduced to no greater than ER, unless ER is less than MCR, in which case ACR shall be set to MCR.

The destination behaviour is also partly described as follows.

- When a data cell is received, the DES shall save the EFCI (per VC).

- On receiving a forward RM cell, the destination shall turn around the cell to return back to the SES. The RM cell fields shall be unchanged except:
 - the direction bit, DIR, shall be changed from 'FORWARD' to 'BACKWARD',
 - if the saved EFCI is set, then the DES shall set CI = 1 in the RM cell and the saved EFCI shall be reset,
 - the DES (having internal congestion) may reduce ER to whatever it can support and/or set CI = 1 or NI = 1,
 - the CRC field shall be recalculated.
- A destination can generate a backward RM cell without having received a forward RM cell. The rate of these backward RM cells shall be limited to 10 cells per second.

Finally, the switch behaviour is partly described as follows.

- The switch shall implement at least one of the following methods to control congestion at queuing points:
 - it may set the EFCI flag in the data cell headers (EFCI marking),
 - it may set CI = 1 or NI = 1 in the forward and/or backward RM cells (relative rate marking),
 - it may reduce the ER field of forward and/or backward RM cells (explicit rate marking).
- A switch may generate backward RM cells. The rate of switch-generated backward RM cells shall be less than 10 cells/sec per VC. When a switch generates an RM cell, it shall set either CI = 1 or NI = 1, shall set the backward explicit congestion notification (BN) = 1, and shall set DIR to BACKWARD.
- RM cells may be serviced out of sequence with respect to data cells. RM cell sequence integrity must be maintained.

2.3 Some features of the SES and DES behaviour

It can be seen that the formal description of the behaviour of the various parts of the control loop is much more complex than given in the oversimplified description of section 2.1. The intention here is to fill in some more of the detail.

First of all, the RM cells include both an explicit rate setting and a congestion indication. It is possible to operate the control loop with only a congestion indication or only an explicit rate. If it is only a congestion indication, the network is relying on the standard behaviour of the SES to lower and raise its rate according to a defined algorithm. Implementing different rate policies for different customers is possible but somewhat inflexible. It has to be done through initialisation of the ABR parameters in the SES.

Another aspect of only using the congestion indication is that the resultant performance is not necessarily fair when multi-hop connections contend with single-hop connections for link bandwidth. There is a tendency for the multi-hop connection to be given much less bandwidth than the single-hop connection, although there is evidence that this is dependent on the exact configuration [3]. The shorter delay in the feedback loop of a single-hop connection may sometimes be a factor in giving either a larger number of step-increases or a faster decline compared with the behaviour of a multi-hop connection.

The explicit rate-only mode can avoid these difficulties. In one variant of this mode, the SES is directly slaved to the latest explicit rate advised by the switches or the DES. This variant is entered if the switches overwrite RM cells so that what is returned to the SES has always $CI = 0$ plus the desired explicit cell rate value. In addition the SES must be initialised so that its step increases are very large — equivalent to going from any rate to at least the physical link rate in one jump. Under these circumstances, according to the sixth rule of the SES behaviour described in section 2.2, the SES will set its sending rate to the least of two values — which will always be the explicit rate, since the other has been step-increased to a large value according to the fifth rule of the SES behaviour in section 2.2.

If the SES is slaved in this way it becomes possible to think of a network controlling element which is responsible for resolving conflicting rate indications within the network and providing a single rate command directly to the SES. These rates may be based on criteria other than strict

fairness. A further discussion of this aspect will be given in section 5.

It is preferable to place such a rate-commanding control entity at the edge of the network as close to the SES as possible — in that way it can be used also as a policing tool, i.e. it sends rate commands and then uses this knowledge to police the sending rate. One idea is to base the controlling entity upon the so-called virtual source/ destination concept. This concept is discussed next.

3. The virtual source/destination (VS/ VD) concept

Referring to Fig 2, an SES is shown connected to a DES, but the control loop is now broken into a number of segments — loop 1, loop 2, and loop 3.

As its name suggests, a VS/VD behaves as if it were the destination of a control loop. Thus, all of the RM cells sent by the SES in control loop 1 terminate at the first VS/VD. This generates backward RM cells towards the SES and it is ideally placed to provide policing at the edge of its network defined by control loop 2.

Some other advantages of the VS/VD are as follows.

- The VS/VD can reduce the rate of sending RM cells across loop 2. It may be noted that 32 is a typical value of N_{rm} (the integer defining the proportion of RM cells — see section 2.2). However, this gives rise to approximately a 6% overhead applying to every ABR VC, i.e. 3% because of forward RM cells and a further 3% because of backward RM cells. Within a public network (e.g. loop 2) this 6% overhead is undesirable and a larger value of N_{rm} may be preferred. A strict reading of the VS/VD behaviour (see below) would suggest it has to send RM cells onwards at the same rate they arrive. However, an RM cell dropping and an RM cell duplication function could be added to the concept such that the original overhead is reduced over loop 2 and then recreated at the boundary between loop

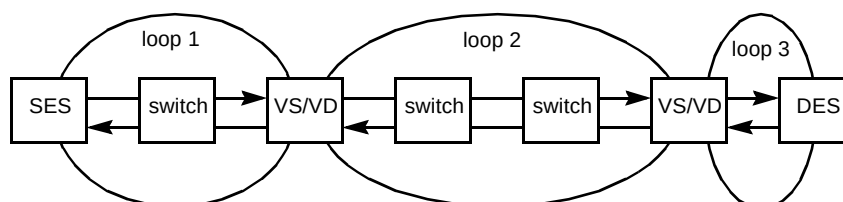


Fig 2 Segmenting the control loop by using a VS/VD.

2 and loop 3. Thus the VS/VD concept provides a way of tuning one of the ABR parameters to optimise performance in WANs and LANs.

- Other possibilities for tuning also exist. Thus it may be observed that the round-trip delay of Loop 1 is small and similar to the delay bound of a LAN. The lower round-trip delay provided by the VS/VD shortens the interval when there is no feedback from the network and this allows either less buffering at the LAN switch or, alternatively, a higher starting rate. It should be noted that, whatever the starting rate of the SES, the VS/VD sets the starting rate policy for its network (e.g. loop 2). Whenever it detects any activity from the SES it can immediately return a command to the SES to go at some other rate, and can set its own output shaper so that this other rate immediately applies within loop 2. Any excess cells would be buffered at the VS/VD.
- The VS/VD can ease the problems of cell delay variation in the control loop. The desired smooth increase in the sending rate is very much dependent on the lack of large cell delay variations. Such variations may cause some RM cells to arrive almost back-to-back. Under these circumstances there may be two or more step increases within a very short period leading to undesirable performance effects. The VS/VD reshapes (smoothes) the flow of control RM cells back to the SES, thus reducing this problem.
- The VS/VD may take account of the fact that the SES has only a limited number of different sending rates. If the VS/VD has a look-up table which contains information about the characteristics of the SES, including minimum and maximum rates and the number of different rates, then it can use this information in its explicit rate-setting policy. It would be necessary to supplement this action with some means of signalling to the other switches in its network (e.g. loop 2) so that unwanted bandwidth is returned to the common pool. One way of achieving this is discussed in the next section.
- A particular policing problem concerns the generation of RM cells which have cell loss priority (CLP) set to 1 in the ATM header. The reason for these is that an SES may have a zero minimum cell rate (MCR) and has returned to this minimum due to prolonged congestion conditions. With a zero minimum cell rate it would not, ordinarily, have permission even to send a forward RM cell that could re-establish some bandwidth. It would therefore be stuck forever with a zero sending rate. To overcome this difficulty the SES is allowed to send up to 10 RM cells per second with CLP = 1. The VS/VD is in an ideal position to intercept these low priority requests and delete them from its network (loop 2) until bandwidth is available. An example behaviour would be that the VS/VD turns around these cells and marks

the explicit rate as zero whilst informing ABR functions in the network of loop 2 (which could be through a proprietary signalling technique) to calculate the bandwidth available for this requesting SES. By deleting the cells at the VS/VD there is no need to support dual CLP priorities in the switches of loop 2.

The VS/VD in control loop 1 can always send backward RM cells with CI = 0, except when it receives a congestion indication from the switch (which may be a LAN hub). In this case it may still mark its own preferred explicit rate, but returns the RM cell with CI = 1. The VS/VD is really performing two functions here:

- to terminate the control loop of a congestion-marking LAN so that this operates as expected,
- to ensure that the SES modifies its dynamic range of sending rates in response to the congestion indication, keeping the VS/VD advised explicit rate as the upper bound.

It must be stressed that it is not necessary to segment the control loop with a VS/VD merely to get two adjacent networks to interoperate if one is congestion-marking only and the other is explicit rate only. There are performance, policy-setting and policing advantages for the VS/VD, as discussed above, but the VS/VD is not a crucial device.

The formal description of the VS/VD behaviour is as follows:

- an ABR connection may be divided into two or more separately controlled ABR segments,
- each ABR control segment, except the first, is sourced by a virtual source, which assumes the behaviour of an ABR source end point — backward RM cells received by a virtual source are removed from the connection,
- each ABR control segment, except the last, is terminated by a virtual destination, which assumes the behaviour of an ABR destination end point,
- the coupling between two adjacent ABR control segments associated with an ABR connection is implementation-specific.

A proposal for a rate-commanding unit based on the VS/VD concept, together with the necessary supporting overlay ABR functionality to support permanent/private VCs (PVCs) or PVPs, is discussed in the next section.

4. Overlaying ABR controls for PVCs/ PVPs

4.1 Overview of the control loop

Why overlay the ABR control loop? The advantage of this approach is that it allows ABR to be introduced sooner on an existing ATM switch platform. Instead of replacing these switches in order to support ABR (which may only be wanted by a small number of customers initially) it creates sufficient controllers to get started. It is also to be expected that ABR will evolve over a period of a few years, e.g. improving allocation policy with respect to single-hop/multi-hop performance, and it may not make good commercial sense to replace existing ATM switches with early ABR versions.

Furthermore, overlaying the control loop is not the same as creating an entirely separate overlay network. The user cells are partially integrated with other non-ABR flows in the same ATM switches. However, it is preferable that ABR user cells do not share the same buffers as used for the DBR, SBR, or ABT capabilities. The reasons for this are given below.

- DBR (and ABT) is expected to use only small buffers. The need to buffer constant bit rate connections is because of the build-up of cell delay variation which causes excess loads, but with very short duration. Typically a cell buffer of 200 cells is adequate to cope with this problem. However, a deeper buffer would be required if it was necessary to cope with both CDV and the fluctuating load produced by ABR. Deep buffers are to be avoided for DBR because it makes the cell delay variation worse, leading to problems possibly with adaptive timing recovery or extra buffering in the receiver to smooth the received flow;
- Since the QOS objective for ABR is very low cell loss, it is impossible to meet this if ABR shares the buffers of SBR, given that the latter is making use of multiplexing to achieve statistical gain. The very fact that statistical gain is employed implies that there will be small intervals of time where cell losses are

significant. Hence ABR cannot be in this mix unless some sort of intelligent cell discard policy is employed, selectively favouring the ABR VCs.

The preference for separate buffering is true regardless of whether the ABR control loops are overlayed or not.

Overlaying the control loop is feasible for PVCs or PVPs but is much more complicated for SVCs (switched virtual circuits) — but some of the disadvantages of PVCs are removed with ABR. The chief one is that bandwidth is not permanently allocated to the VC, even when there is no activity. ABR controls remove this bandwidth and the customer only gets a further allocation when next active. The chief remaining disadvantage is the problem of scaling network management operations to cope with a large number of PVCs. Nevertheless it is suggested that a healthy market for PVCs is expected to continue and ABR PVCs should be part of the portfolio of service offerings.

Referring to Fig 3, two ABR PVCs are shown entering ATM switch 1 which crossconnects them to a device known as the dynamic bandwidth controller (DBC). The purpose of the DBC is to act like a VS/VD in that it provides RM feedback to the SESs. It additionally provides policing, buffering, re-shaping, and rate control functions [4].

From the DBC the PVCs re-enter switch 1 and then continue on to other ATM switches which are each monitored by an ABR rate control function (which will be explained below). The user flow is not diverted through the rate control functions — they act only to support the control loop.

The DBC is in contact with the rate controllers on adjacent switches through the establishment of control PVCs, as shown in Fig 4.

This configuration allows the DBC to send a non-standardised RM cell along control path 1—2 to the rate controller on switch 2 whenever it detects activity or lack of activity on particular VCs. The VCs in question are the ones that are routed from switch 1 to switch 2. There may be

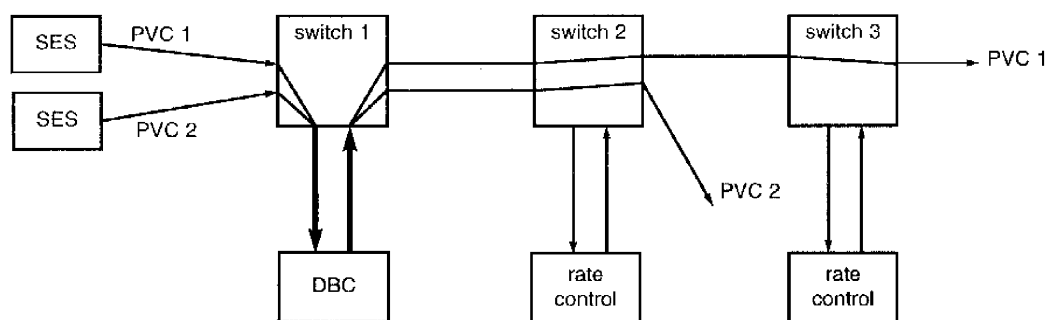


Fig 3 Proposed overlay of the ABR control loop.

other switches adjacent to switch 1. VCs routed from switch 1 to those switches will not require any control signal to be sent along control path 1—2.

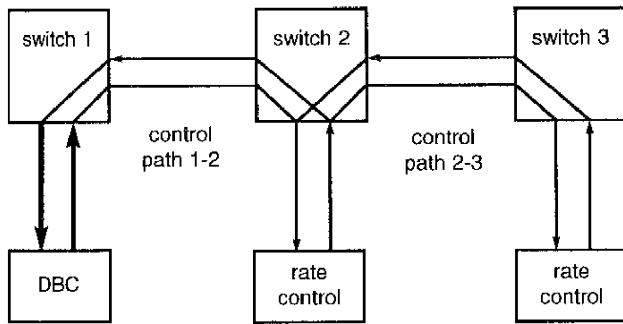


Fig 4 The routing of control information between the DBC and the rate controllers.

The main purpose of a rate controller is to keep a log of all active VCs on a given output port of its co-located switch and to advise the bandwidth allocation per VC. For example, the rate controller on switch 2 keeps a log of activity on all of the switch ports of switch 2. If a new VC becomes active, the rate controller will look up which switch port it uses on switch 2 and then identify the currently available bandwidth (using the algorithm described below in section 4.2). It sends a signal back to the DBC when it has resolved the maximum available bandwidth with other rate controllers on the route of this VC.

The DBC acts as the rate controller for its own co-located switch.

4.2 Rate allocation

4.2.1 Finding out which VCs are contending with each other

The log of activity within rate controllers is built up only from messages appearing on control paths, since these controllers do not directly receive/forward data cells on any ABR VC.

The rate controllers must derive a map of which VC uses which switch output port, so that they can correctly determine the available bandwidth. There are several possibilities here. Self-learning techniques are to be preferred because they scale more easily than using network management methods, and are able to deal quickly with some failure conditions which require automatic re-routing of the VCs.

Self-learning is not possible in one area. At the DBC there is nothing in the standard which allows the DBC to discover the identity of the SES from its incoming RM cells. Therefore the map between the customer identity and VPI/

VCI value of the PVC entering or exiting the DBC must be provided by network management when the connection is established. The rest can be self-learned.

Self-learning is possible by making use of RM cells sent forward from the DBC along the user-path. If switches can be configured so that these RM cells are diverted to the rate controller on arrival at the input port¹, then routing information can be gradually built up. Thus, as an RM cell arrives at a rate controller, it returns an RM cell along the backward path of that VC with a field which specifies the rate controller's own address. For convenience, let us think of that address as, for example, 'switch 2'. When the DBC receives this information it now knows that the VC must use the output port of switch 1 that corresponds to the route towards switch 2. The process is repeated by the rate controller at switch 2 sending a forward RM cell along the user path, and getting a backward RM cell from, for example, switch 3. This process requires that the switch internally routes some RM cells forwards and others backwards as they arrive from a rate controller, e.g. at the input port R of Fig 5.

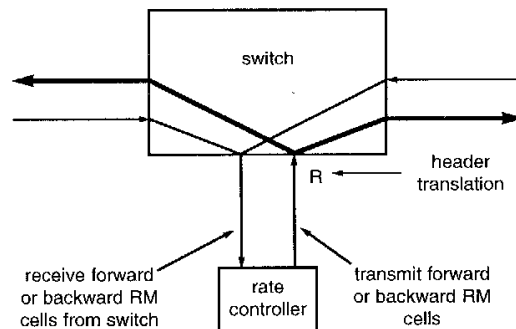


Fig 5 Part of the control loop.

It is not necessary for the input port of the switch to look at the information field of the RM cell to discover whether it should be routed forwards or backwards. A simpler convention would be for the rate controller to use one bit of the VPI field to signify forwards or backwards together with header translation at R so that the RM cell is correctly re-labelled before being internally routed to join the user flow.

The use of one bit of the VPI field implies that this limitation is known by call acceptance control when it assigns VPI/VCI values.

4.2.2 The 'Robin Hood' algorithm

The simplest type of algorithm is that a newly active VC is given just one PCR which is retained until the connection goes quiet again. Only when it subsequently reactivates will it get a different PCR.

¹ If it is not possible to configure switches this way, then self-learning is only achievable if the rate controllers are designed to receive the user flows in addition to the RM cells.

Furthermore, putting it simply, a 'now gone inactive' signal from the DBC relating to one VC will cause no control signals to be generated on the other VCs which were sharing capacity with it. Instead, the capacity it was using is now re-labelled 'spare'.

This must be combined with a filling algorithm. For example, one filling algorithm is:

- to give the first newly active connection an allocation which is $\frac{1}{2}$ of the total available capacity — this is idealistic where every SES is assumed to be capable of using whatever bandwidth is made available,
- the next newly active connection is given an allocation which is $\frac{1}{2}$ of the remaining capacity,
- the next newly active connection is given an allocation which is $\frac{1}{2}$ of the still remaining capacity, and so on.

This algorithm must be applied link by link over the entire route of the VC, and whichever yields the lowest allocation is the determinant of the PCR sent back from the DBC.

The main advantage with this simple algorithm is that a 'newly active' signal for one VC generates no allocation changes on the other VCs which are sharing capacity with it. This minimises the number of control actions to be performed.

This algorithm is fair in the sense that, over a long enough period, no customer is systematically given a poorer capacity. It is also expected to be the easiest to implement in a large public network. However it is desirable to increase the number of customers who get largish capacities and this leads to the modified algorithm called 'Robin Hood', which is now discussed.

The basic principle here is that, if an active signal causes control signals on other VCs, let it be limited to only one per link, namely the richest (largest capacity) VC.

This can best be illustrated with an example filling algorithm:

- the first newly active VC is assigned an allocation equal to $\frac{1}{2}$ of the total available capacity,
- the next newly active connection is assigned $\frac{1}{2}$ of the remaining capacity plus $\frac{1}{5}$ of the allocation of the first VC, i.e. the current richest,
- the next newly active connection is assigned $\frac{1}{2}$ of the remaining capacity plus $\frac{1}{5}$ from the current richest, and so on.

To illustrate this process, imagine there is just one link with a capacity of 100 Mbit/s.

- the first VC gets 50 Mbit/s, and there is 50 Mbit/s remaining,
- the next VC gets $\frac{1}{2}$ of the remainder (which yields 25 Mbit/s) plus a $\frac{1}{5}$ from the first, which therefore, now has 40 Mbit/s, with the second now having 35 Mbit/s,
- the next VC gets $\frac{1}{2}$ of the remainder (which yields 12.5 Mbit/s) plus $\frac{1}{5}$ from the first, which, therefore, now has 32 Mbit/s, the second still having 35 Mbit/s, and the third 20.5 Mbit/s, and so on.

It should be noted that more of the customers are now getting larger capacities but there is only one extra control message to send on this link.

To extend the algorithm to a route with many links the process is as follows. The above process is repeated link-by-link. Whichever link yields the lowest allocation is the determinant of the PCR sent back to the DBC. Now, using this value of allocation, the DBC informs the rate controllers (i.e. it informs the adjacent rate controller, who then informs the next, etc) to assign this allocation. Each rate controller does this by taking up to $\frac{1}{2}$ of the remaining capacity on that link (if needed) and any extra which is needed is taken from the richest VC on that link. So this generates at most one additional PCR control message per link for each VC active signal sent to the network. A 'now gone inactive' signal still generates no additional control messages.

This strategy also makes it impossible for a customer to hold on to a very large capacity when others become active. It seems to satisfy the desire to get as many customers as possible a fairly large capacity while keeping the complexity of rebalancing to a minimum.

4.2.3 Determining which link has the least capacity

It has already been mentioned that the DBC becomes informed of the PCR which is of least value, derived from all of the rate controllers along the path of a VC. How is this done?

Across the public network a VC starts and ends on a DBC. When the far-end DBC (farthest from the SES) receives a 'newly active' signal it knows that there are no other rate controllers beyond it and so it immediately sends back an RM cell to the next adjacent rate controller along the backward path. The RM cell contains the PCR value which this DBC can allocate merely looking at the bandwidth available at the receiving customer's user-network interface. It must be remembered that network management has told the DBC the association between VPI/VCI value and customer identity. Similarly, network

management will have told the DBC about the maximum bandwidth available for ABR on the DES.

At a rate controller, following receipt of a 'newly active' signal (naming the VPI/VCI), no RM cell is sent back until firstly an RM cell is received in the backward direction. This implies that the far-end DBC will be the first to send an indication backwards, then the next adjacent rate controller, then the next, and so on. As each rate controller sends an indication backwards it writes into the RM cell the minimum of the PCR determined from its own algorithm and the PCR arriving on the backward RM cell. By this process the DBC gets informed of the maximum acceptable rate.

All of this takes some time, so when the DBC detects new activity it does not wait, but immediately informs the SES to go at some minimum or default rate (which can be set differently for different customers). It shapes the output flow to this rate and buffers any excess cells.

4.3 Overview of the DBC

A brief overview is given here, and a detailed description can be found in Adams and Smith [4]. Referring to Fig 6, the DBC has the following components — activity detector, per-VC buffer, controller, shaper/scheduler, feedback.

- Activity detector

As its name suggests, the activity detector senses new activity or cessation of activity on each VC and informs the controller of these events. It also keeps a running check on the utilisation of assigned bandwidth when required to do so by the controller. Significant under utilisation results in a warning signal to the controller.

- Per-VC buffer

This acts as part of a policer (together with the shaper/scheduler). It provides enough buffering to compensate for the round-trip delay of the rate commands, so that the shaper/scheduler can be immediately set to the latest desired rate.

- Controller

The controller is the heart of the DBC and includes a rate control algorithm so that the DBC can act as a local rate controller. The controller is responsible for triggering the feedback module to send rate commands to the SES (through RM cells) and for generating forward RM cells at the scheduler which will be sent towards the next rate controller. It receives backward RM cells from the feedback module, and uses this information to adjust the VC sending rates.

- Shaper/scheduler

This extracts cells from the per-VC buffer and forwards them at the desired rate (which includes some level of CDV tolerance). Cells arriving from a single SES which are meeting the acceptable limits on CDV tolerance will not be delayed at all by the shaper/scheduler if there are no cells waiting in the buffer from another SES. The scheduler manages the problem of several contending SESs whose flows are to be output from the DBC.

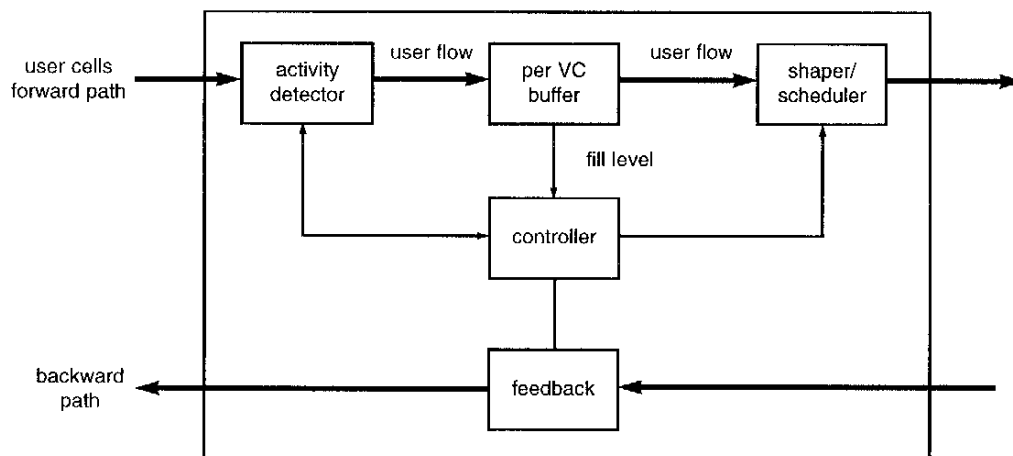


Fig 6 The DBC.

- Feedback

The feedback component formats feedback information in RM cells and transmits them towards the SES. The feedback information is provided to this component from the controller. A second function of this component is to remove and unpack RM cells sent towards the DBC from a rate controller, and then forward the information to the controller.

5. Extension of the ABR service concept

The control system described in section 4 is complex and only applies to PVCs or PVPs. It is necessarily complex because of what it is trying to do. It is controlling individual bursts of data activity to such an extent that they almost never cause congestion, while still allowing very high utilisation of network resources. In every case when a burst commences, the nearby DBC returns an immediate command to go at a very slow speed. Then, after the control loop has completed its process (in a period of, perhaps, 30 ms or less depending mainly on the propagation delay), a single step increase in rate is commanded by the DBC. At this point, the end-system is typically enjoying a high sending rate and, through the Robin Hood algorithm or variants of it, can expect relatively few changes of rate.

How may such a system be expected to be used, and what extensions of the basic concept are possible? First of all, why bother at all? Surely TCP can do all this. However, TCP works best in an environment where congestion is infrequent. If the maximum sending rate of the SES is very small compared to the link capacities in the network, it is obvious that TCP will do a good job. The SES will increase its rate to the maximum through the TCP algorithm but still will not experience congestion on most occasions. There is no need for all of the above complexity in this case.

The problem is that the ideal scenario just described is not the one expected to be true in the very near future. It is likely that there will be SESs with very high sending rates compared to the network link capacities. Under these circumstances just a few SESs can cause overload and, without other controls, the frequency of entering congestion will be unacceptable, causing throughput collapse.

A traditional way of dealing with this problem is overdimensioning of the networks. Typically the network planner will project traffic load trends some way into the future and provision on the basis of meeting low congestion frequency throughout that period. However, this may be an unacceptable cost, at least over the next decade (perhaps later on, capacity will be less cost sensitive, e.g. if an all-optical network — switching as well as transmission — becomes a reality).

Thus a window of opportunity — a decade or more — seems to be open for ABR. Its job is to deal with high-rate end systems contending for limited network capacity. How may a PVC/PVP service be expected to be used in this respect?

Referring to Fig 7, a LAN emulation service is shown which involves two sites linked by an ABR PVP.

It should be noted that, in Fig 7, some licence has been taken in drawing the VP and VCs to show the multiplexing arrangement more clearly. In practice, each end of the VP will terminate directly on a hub.

The hubs may be able to support SVCs, and all such VCs that are routed from one hub to the other can be assigned the same VP value, i.e. at the exit point from the hub towards the public network. This still allows approximately 4000 simultaneous connections between the hubs over the VP, in addition to the connections which remain within one hub site. If this address space is not big enough, then two or more ABR VPs could be used and differently routed for added gains in security and throughput.

The ABR PVP does not look at the individual VCs, but only at the total load. It is unaware of VCs being set up and cleared down. One of the VCs is a signalling channel that enables all of the SVC operations to be co-ordinated between the two hubs. No signalling is terminated in the public network.

The LAN emulation service relies on a server function that, as one of its jobs, facilitates the sending of ATM addresses to the SESs which then initiate calling procedures

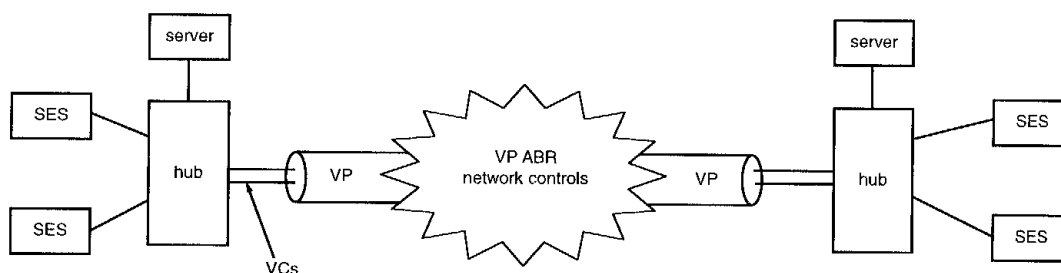


Fig 7 LAN emulation supported by ABR VPs.

to set up a connection for the duration of a session. Only when the hub receives these calling indications will it establish new VCs across the PVP and co-ordinate their correct termination at the far end through signalling to the next hub. An important property of LAN emulation is that these new VCs should not be refused admission on the basis of lack of capacity. This means that the VCs will sometimes be very slow but will always be capable of some throughput (in this case down to a minimum of approximately 1/4000 of the VP capacity).

What the ABR controls are providing is a method of maximising the VP capacity with very few cell losses. Equally the ABR controls are removing unwanted capacity from VPs which are quiet. This represents an intelligent bandwidth adjustment 'service' on a private connection between two LAN sites.

Congestion can still occur at the ingress and egress of the VP. At the ingress, congestion occurs because the individual VCs are sending more load than can be carried away by the VP. The result is that a buffer starts to fill at the near-end hub. There are two alternative possibilities.

- The VCs may also be controlled through ABR. The VC-level ABR control need only be present in the private networks, not in the public network. Thus, as a hub buffer starts to fill, it marks the EFCI forwards or changes explicit rate settings on the individual VCs. They pass transparently through the VP, which is only serving to back pressure the VCs with its capacity control. The returning backward RM cells on the VCs cause the SESs to back off. This is a very good solution to both ingress and egress congestion.
- The VCs are uncontrolled. Only at the transport layer is there any mechanism for adjusting the flow rates. Even in this case the ABR VP is providing some benefit — by confining the interference to other VPs when there is congestion at the ingress of one VP. It is like an intelligent cell discard policy which selectively targets some VCs while leaving others unaffected. The main aim must be to impose a bandwidth allocation policy such that, on the average, moments of congestion are just as frequent (but not usually co-incident) at each of the VP ingresses. Then the intelligent discard policy is also fair.

Therefore the problem that needs addressing is the VP bandwidth allocation policy. The solution is a development of the control loop that is currently not part of the standard — implementing feedback between a policing function and the allocation control.

Unlike VCs, it may be expected that activity is very prolonged on the VPs. One or more VCs in any VP will be sending data, so moments of true quietness are infrequent.

The allocation control policy must therefore have a different aim — to continually adjust bandwidth so that:

- if too much bandwidth has been assigned it is removed, so that other VPs can benefit,
- if all the bandwidth that has been assigned is being used, the VP is identified as a candidate for an increased allocation. Assuming this aim can be met, it implies that the VP allocation policy is approximately fair in the intelligent discard policy sense described above. It should be noted that it is not fair in the sense of a weighted allocation policy — the concept that some VPs are given a predefined larger allocation than others. A typical weighted allocation policy would define weights for a total of S contending VPs as follows:

let w_i be the weight assigned to VP_i , such that

$$\sum w_i = 1, \quad \text{where the summation is over all } S \text{ VPs;}$$

now suppose only R of the S VPs are active; then the bandwidth allocated to VP_i is:

$$\text{Allocation}_i = \text{Link Capacity} * w_i / (w_1 + w_2 + \dots + w_R)$$

The trouble with the weighted allocation policy is that it blindly assigns bandwidth to a VP even if some of that bandwidth remains unused. The converse argument is that the 'intelligent-discard' policy above favours VPs with only a few active VCs who each grab a large rate. There may be no ideal compromise between these two positions that is not very difficult to implement (remembering that it should not be necessary to be aware of the individual VCs that have become active if the control is at the VP level only). This is an area for further investigation.

How do the DBC/rate controllers implement an intelligent discard policy? The controller in the DBC switches on the activity detector to monitor for under-utilisation. If this condition is detected the DBC immediately informs the VP source (typically, a hub) to go at a slower rate. It also sends an RM cell forward to the next adjacent rate controller along the route to release the excess bandwidth. The rate controller keeps a record of how much unused bandwidth exists — it does not attempt to re-assign it until it gets another instruction from a DBC. Having done this the rate controller also sends an RM cell forward to the next adjacent rate controller, informing it to do the same, and so on.

The DBC will send an RM cell forward requesting more allocation for a VP, if under-utilisation has not occurred for a given period. If more than one candidate exists in this category, it selects them in a round-robin fashion so that the frequency of sending requests forward is limited to, perhaps, no more than one per second. When the rate

controller receives this request it assigns some of the bandwidth in the free pool plus some bandwidth from the richest VP in the Robin Hood fashion. Thus VPs which are currently bandwidth rich are not guaranteed to hang on to their allocation by keeping the sending rate high.

This completes the description of ABR VPs supporting a LAN emulation service. An additional feature is that a VP level of control significantly improves the network management task. The network is not responsible for setting up and tearing down individual VCs — this task is left to SVC operations in the hubs.

6. Conclusions

The concept of ABR has been described, together with an introduction strategy that applies it to private virtual paths which have an intelligent bandwidth allocation control. It has been shown how such VPs can be used to support SVC operations for LAN emulation, where the VC level may or may not have any ABR capability.

The main conclusion to emphasise is that control concepts are still being refined and flexibility in setting allocation policies is very important. For instance it would be nice to look at other extensions of the allocation policy not already mentioned in this paper — including VCs where a large allocation (say, 80 Mbit/s) is made available only when the network is quiet; otherwise no allocation is given. Such VCs could be used for performing data back-ups at a time when the network has bandwidth which it cannot otherwise sell.

Furthermore there will be a need for lengthy study of the best settings of ABR parameters to optimise the performance in both LANs and WANs, taking account of the interaction of ABR with the transport layer.

The proposed introduction strategy is to place the intelligent ABR control functions external to the switches so that services can be tried out more rapidly. It is recommended that there is no substitute for trying the functionality out in a live traffic environment — one which allows a limited number of customers to participate and confines the extent to which they can interfere with other traffic. The DBC/rate-controller environment fits exactly with this strategy.

In implementing this strategy, a preference exists for switches which can place ABR traffic in a separate buffer to the other traffic types. This implies at least three buffers per

output port if constant bit rates and non-ABR statistically multiplexed traffic are included. A preference also exists for switches that can separately route the RM cells within a VC or VP. This allows the intelligence to become external to the switches.

A private VP-level ABR service is seen as being not too demanding in terms of network management operations. There may be thousands of individual VCs within each VP (many or most of which are quiet at any one time). The responsibility for establishing these VCs lies with the SVC controls implemented in the data hubs. At the VP level in the public network these operations are not seen.

One aside is that communities of interest are normal in the data communications world, so that a very large percentage of the total load sent from any one customer site is going to just a few other sites. This picture fits well with permanent VPs. It is only necessary to have a fully switched public service to deal with the much smaller fraction of traffic that is going to infrequently used destinations, where a permanent VP would be inappropriate.

Finally, some speculation. The first specification of an ABR standard is expected to be completed in the latter half of 1995 — but it is likely to be only the beginning, because it seems evident that the concept is a very powerful one. Having end systems which are slaved (in terms of sending rate) to a feedback loop opens the door to all sorts of possibilities. Some people are already talking of ABR as the next new technology, as once people spoke about ATM. In saying this they are visualising ABR-type controls for video, voice, and data — time will tell if they are right.

Appendix

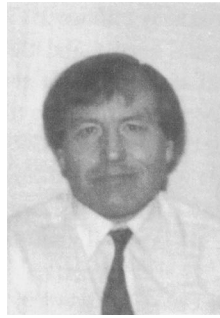
List of acronyms

ABT	ATM block transfer (capability)
ABR	available bit rate (capability)
ACR	allowed cell rate
ATM	asynchronous transfer mode
BN	backward explicit congestion notification cell
CDV	cell delay variation
CI	congestion indication
CLP	cell loss priority
DBC	dynamic bandwidth controller
DBR	deterministic bit rate (capability)
DES	destination end system (e.g. a router)
EFCl	explicit forward congestion indication
ER	explicit rate

ICR	initial cell rate
LAN	local area network
MCR	minimum cell rate
NI	no increase
PCR	peak cell rate
PVC	private virtual circuit
PVP	private virtual path
QOS	quality of service
RM	resource management
SBR	statistical bit rate (capability)
SCR	sustained cell rate
SES	source end system (e.g. a router)
SVC	switched virtual circuit
TCP	transmission control protocol
VS/VD	virtual source/ destination

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